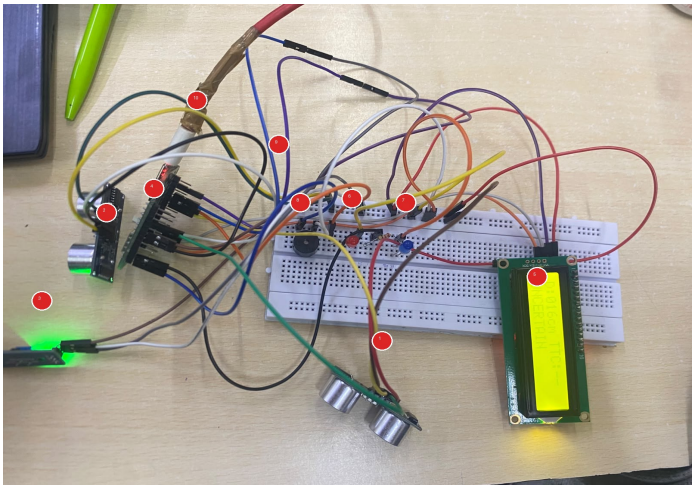
DOCUMENT 3 OF 5

BUILT HARDWARE, ANNOTATED PHOTOS

Physical Hardware Prototype Bench Examination, Module Callouts and Workmanship Flags



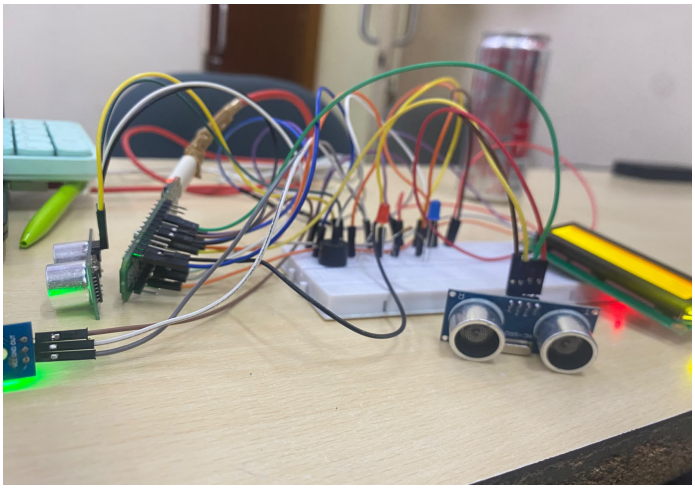


Figure 3.1 (Left): *Top-down bench photograph with numbered callout tags.* ***Figure 3.2 (Right):*** *Eye-level perspective photograph verifying orthogonal sensor orientation.*

Numbered Component Callout Legend

Callout #	Subsystem / Component	Observed Hardware State & Verification Note
`[1]`	HC-SR04 Lateral Sensor (B)	Mounted facing perpendicularly across traffic flow; provides blind-spot cross-traffic ranging.
`[2]`	HC-SR04 Front Sensor (A)	Mounted facing forward along longitudinal axis; tracks approaching vehicles from 2 cm to 100 cm.
`[3]`	TCRT5000 IR Contact Sensor	Ground-truth bumper impact detector; active green power LED confirmed illuminated on board.
`[4]`	ESP32-H2 Dev Module (Glyph H2)	Core microcontroller running at 96 MHz; power LED illuminated, USB-C connected.
`[5]`	16x2 Character LCD Display	I2C yellow-backlit display actively rendering verified text: 'D: 016cm TTC: UNCERTAIN'.
`[6]`	Red Hazard Alert LED	5mm diffused red LED wired in series with 220Ω resistor on breadboard rows c9-c13.
`[7]`	Blue Monitor LED	5mm diffused blue LED wired in series with 220Ω resistor on breadboard row 17.
`[8]`	Active Piezo Buzzer	Standard 12mm cylindrical active buzzer inserted on row 5 for audible hazard alert beeping.
`[9]`	Central Jumper Wire Bundle	High-density flying Dupont jumper wires distributing 3.3V, GND, and GPIO logic signals.
`[10]`	USB-C Interface Cable	Provides 5V VBUS power and establishes 115200 baud USB CDC telemetry stream to dashboard.

Workmanship & Quality Inspection Flags

Flag ID	Severity	Inspection Item	Observed Defect & Recommended Corrective Action
`FLAG-01`	MODERATE	Central Wire Bundle Congestion	Flying jumper wires form an unbundled cluster over the center breadboard rows. Obscures pin contact inspection. Recommendation: Loom wires with spiral wrap or zip-ties.
`FLAG-02`	HIGH	USB-C Cable Strain Relief	Brown adhesive tape is wrapped around the USB-C cable boot entering the ESP32-H2 board, indicating cable fatigue or loose socket retention. Recommendation: Use molded strain relief clamp.
`FLAG-03`	MODERATE	Uninsulated Component Leads	Axial leads of 220Ω resistors and LED anodes stand exposed above breadboard row 13. Risk of bridging if bumped. Recommendation: Slide heat-shrink tubing over bare leads.